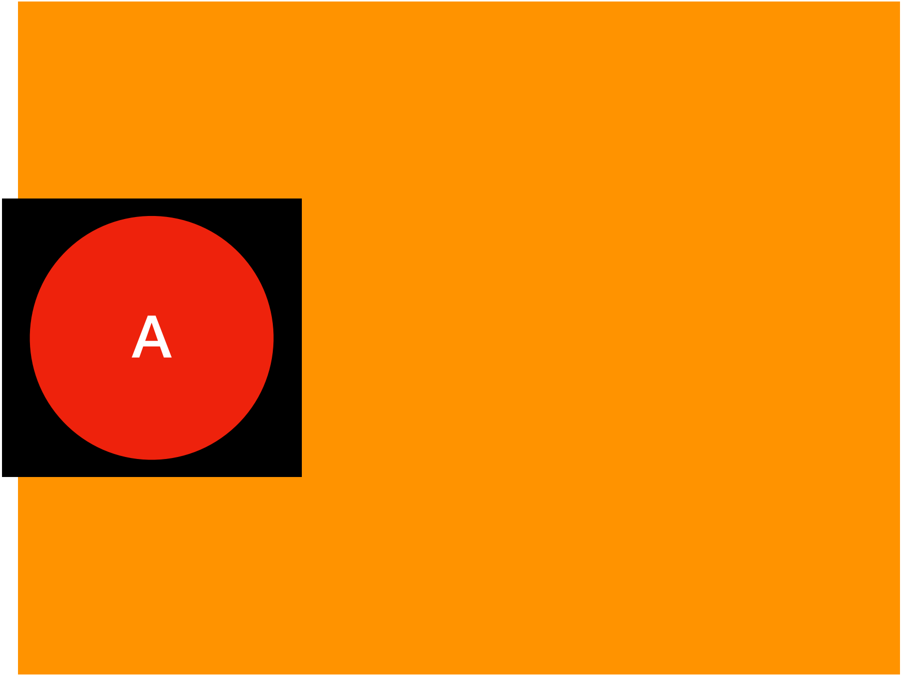

A

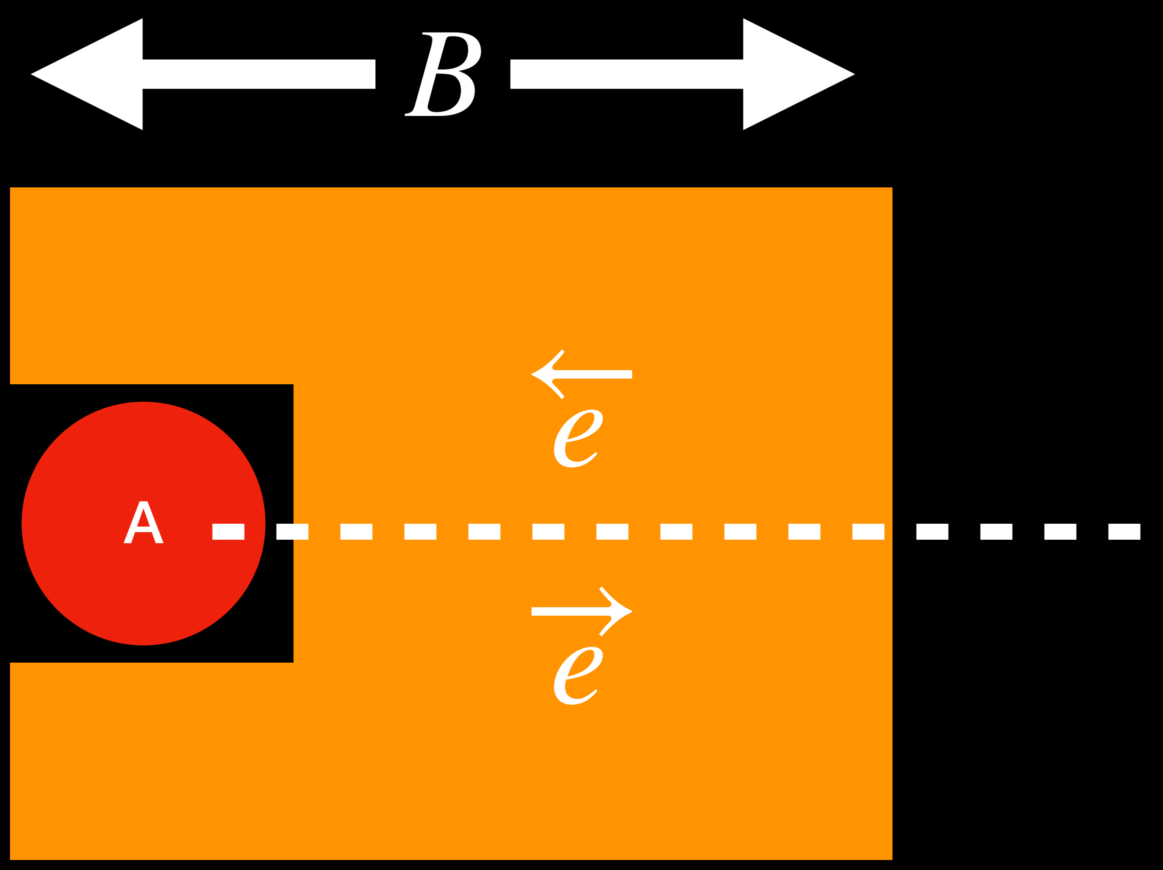
How can we measure $S_z(\gamma)$?

$\overrightarrow{\gamma} + \overleftarrow{e} \to \overrightarrow{e}$ Larger absorption

$\overrightarrow{\gamma} + \overrightarrow{e} \to \overrightarrow{\;e\;}$ Smaller absorption

$$\boxed{\frac{d\sigma(\downarrow\uparrow)}{d\Omega} > \frac{d\sigma(\uparrow\uparrow)}{d\Omega}}$$

A refined statement



Measure:

$N_\gamma^{\text{trans}}(\overrightarrow{B})$

$N_\gamma^{\text{trans}}(\overleftarrow{B})$

$S_z(\gamma) = +1$ if

$$N_\gamma^{\text{trans}}(\overrightarrow{B}) < N_\gamma^{\text{trans}}(\overleftarrow{B})$$

$S_z(\gamma) = -1$ if

$$N_\gamma^{\text{trans}}(\overrightarrow{B}) > N_\gamma^{\text{trans}}(\overleftarrow{B})$$

There is a problem…